

Reproducing the Gu 2022 GIGA-Lens Fast-Bayesian Strong-Lens Modeling Benchmark

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Abstract

We reproduce the simulation-recovery benchmark of [Gu et al. \(2022\)](#) (GIGA-Lens; ApJ 935, 49) end-to-end with the upstream `gigalens` JAX code. Twelve EPL + external-shear lenses with elliptical-Sérsic lens and source light are simulated self-consistently on the paper’s grid (80×80 at $0.065''/\text{px}$, $k_{\text{super}} = 2$, TinyTim F140W PSF, median integrated arc SNR ≈ 116) and fit with the paper’s three-stage pipeline: 128-start MAP, a Gaussian-surrogate SVI step, then preconditioned Hamiltonian Monte Carlo (50 chains) with gradient-based trajectory-length adaptation, dual-averaging step-size adaptation, and mass matrix $\tilde{M} = \Sigma_{\text{VI}}^{-1}$ — matching the paper’s Table 1 settings. MAP recovers the injected truth at reduced $\chi^2_{\nu} \approx 1.0$ (range 1.01–1.11) on *all* 12 systems. On well-conditioned systems we *meet* the paper’s convergence targets (e.g. systems 4/8/9/10: min-ESS 7,500–18,000, max- $\hat{R}$ $1.004\text{--}1.015 \leq$ the paper’s 1.017), with $\mu_z \approx 0$ confirming unbiased recovery. The pooled worst case (min-ESS 71, max- $\hat{R}$ 5.97) falls short of the paper’s *uniform* min-ESS = 26,822 / max- $\hat{R} \leq 1.017$ across all 100 systems; the large μ_z values track exactly the low-ESS/high- $\hat{R}$ parameters — a non-convergence (under-estimated-variance) artifact, not a bias. A depth test re-running the three worst systems at $2.5\times$ chain length roughly doubles min-ESS and halves excess $\hat{R}$ monotonically and without bias. We conclude the GIGA-Lens *method* is faithfully reproduced; the residual gap to the paper’s uniform high-ESS target is a sampling-depth limitation of running degenerate strong-lens posteriors on a single 15 GB A16 (the paper used $4\times$ A100), consistent with the Foundry I ([Huang et al., 2025](#)) HMC-degeneracy finding rather than a method defect.

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1 Introduction

Gu et al. (2022) introduce GIGA-Lens, a GPU-accelerated, fully differentiable framework for Bayesian strong-gravitational-lens modeling. Where traditional samplers (e.g. `emcee` as configured by `lenstronomy`; Birrer et al. 2021) take hours per system, GIGA-Lens models a single lens in $\sim$ minutes by combining JAX (Bradbury et al., 2018) automatic differentiation, TensorFlow Probability (Dillon et al., 2017) inference primitives, and a three-stage optimize-then-sample pipeline that runs many chains in parallel on a GPU. The paper reports an average time of 105 s per system on a 4-GPU NERSC Perlmutter A100 node (6 min on a single A100 for the reference system).

The paper’s headline validation (§3, Figs. 10–11, Table 2) is a simulation-recovery benchmark: simulate **100 mock lenses** with `lenstronomy`, recover their parameters with the GIGA-Lens MAP $\rightarrow$ SVI $\rightarrow$ HMC pipeline, and demonstrate that for *every* one of the 22 model parameters over *all* 100 systems the convergence diagnostics satisfy a uniform target. Quoting the paper directly: “even the largest $\hat{R}$ for any parameter over all 100 simulated systems was 1.017, and the smallest ESS was 26,822, an order of magnitude higher than the typical value with `emcee`.” The per-system mean ESS sits near 32,000 (Table 2 row values 35,382–35,777 and 26,822–33,752), and the posterior means are consistent with the injected truth (mean scaled error $\mu_z \approx 0$). This report reproduces that benchmark on a 12-system validation subset.

Why this benchmark, and the relation to Foundry I. GIGA-Lens is the modeling engine of the DESI strong-lens program: it is the tool used in Huang et al. (2025) (Foundry I) to model HST imaging of candidates discovered in the DESI Legacy Surveys (Huang et al., 2021). The Foundry I reproduction (a companion tech-report) found that, on *real* degenerate lens posteriors, single-device HMC mixing is the binding constraint — the same phenomenon we quantify here on *controlled* mocks with known ground truth. Reproducing the Gu+2022 benchmark therefore both validates the upstream code we rely on downstream and isolates the sampling-depth behavior from the modeling correctness.

Scope. We reproduce the *method* and its convergence behavior on a 12-system validation subset, not the full 100-system Table 2 on identical hardware. The 12 systems are simulated and fit self-consistently within the `gigalens` forward model (see §2.1); the model, grid, noise, priors, and HMC kernel stack are the paper’s. The deltas from the paper — single 15 GB A16 vs. 4 $\times$ A100, reduced MAP multistart and SVI MC batch to fit memory, float32 — are documented in §5.

2 Methods

The pipeline is implemented in five numbered scripts at `reproductions/gu-2022/`: `01_gen_mock.py` (simulation), `02_fit_system.py` (one-system MAP $\rightarrow$ SVI $\rightarrow$ HMC), `03_run_batch.sh` (GPU-cycling batch driver), `04_pool_results.py` (Table-2 pooling), and `05_recovery_plot.py` (Fig.-11-style recovery plot). The `gigalens` source is at `/raid/benson/lensing-repos/gigalens`; the environment is `/raid/benson/.venvs/gigalens` (JAX 0.6.2, TFP 0.25.0, `lenstronomy` 1.14.0, Optax).

2.1 Mock generation

Each system is an EPL mass profile (Tessore and Metcalf, 2015) plus external shear, an elliptical-Sérsic lens light, and an elliptical-Sérsic source light, drawn from the simulation distribution of the paper’s Eq. (8) (the “*a*” side of its *a/b* prior notation; the broader “*b*” side is the modeling prior in §2.2). The grid and noise constants reproduce the Fig. 10 caption and §2.2: pixel scale $0.065''/\text{px}$, 80×80 cutout ($5.2'' \times 5.2''$), supersampling factor $k_{\text{super}} = 2$, $\sigma_{\text{bkg}} = 0.2$, $t_{\text{exp}} = 100$ s, gain $G = 1$ (HST WFC3/F140W-class), and the GIGA-Lens TinyTim PSF kernel (`gigalens/assets/psf.npy`). Noise is Gaussian background plus Poisson shot noise.

Self-consistent simulation (a methodological prerequisite). We simulate with the *same* `gigalens` forward model the fitter uses, rather than `lenstronomy`. At $k_{\text{super}} = 1$ the two codes agree at the identical truth to $\chi^2 = 0.000$, but at the paper’s $k_{\text{super}} = 2$ they differ ($\chi^2 \approx 0.94$ at identical truth) because their supersampled-PSF-convolution conventions (`lenstronomy` `subgrid_kernel`+supersampling-convolution vs. `gigalens`’s internal supersampling) are not bit-identical. Fitting a `lenstronomy` mock with `gigalens` then leaves an irreducible $\chi^2 \approx 0.94$ floor that biases the recovered γ by $\sim 16\%$. Simulating with the same forward model removes the floor (MAP reaches the noise floor $\chi^2_{\nu} \approx 1.0$ with unbiased parameters); the paper’s own benchmark simulates and infers within one framework, and we do the same. The 12 systems have median integrated arc SNR ≈ 116 (range 45–453), bracketing the paper’s ~ 100 target.

2.2 Three-stage pipeline

The fitter (`02_fit_system.py`) runs the paper’s three stages, each with the Table-1 hyperparameters of the reference system, on *one* mock pinned to *one* GPU.

1. **MAP.** Multi-start gradient descent in the unconstrained parameter space: $K_{\text{MAP}} = 128$ random starts (paper 300; see §5), 250 Adam/AdaBelief steps with learning rate annealed $10^{-2} \rightarrow 10^{-3}$. Output: the best- χ^2 optimum.
2. **SVI.** A full-rank Gaussian surrogate posterior $q(\tilde{\Theta})$ is fit by stochastic variational inference initialized at the MAP, 500 ELBO steps, $K_{\text{VI}} = 200$ Monte-Carlo samples/step (paper 1000), initial scale $\Sigma = 10^{-3}I$. This both refines the point estimate and yields the covariance Σ_{VI} used to precondition HMC.
3. **HMC.** 50 chains initialized by sampling from q , run with TFP’s `PreconditionedHamiltonianMonteCarlo` ($L = 5$ initial leapfrog steps, $\epsilon_0 = 0.3$) wrapped in `GradientBasedTrajectoryLengthAdaptation` (the ChEES-style adaptive trajectory length of Hoffman et al. 2021) and `DualAveragingStepSizeAdaptation` (step-size adaptation of Hoffman and Gelman 2014), both adapting over the first 80% of the 250 burn-in steps, then 750 retained draws. The momentum covariance is $\tilde{M} = \Sigma_{\text{VI}}^{-1}$ (preconditioned HMC, exactly the paper’s choice). Diagnostics are cross-chain split- $\hat{R}$ and ESS per parameter.

Single-device GBTLA recipe. The 50 chains are run as a single *batched* `sample_chain` state of shape (50, 22) on one pinned GPU, *not* via the `gigalens` `HMC()` helper. That helper `jax.pmaps` the chains over all visible devices and hangs on a multi-GPU A16 node; the single-device batched formulation is functionally identical (same kernel stack, same adaptation, same $\tilde{M}$) but deterministic to launch. This avoidance of the `pmap`-over-all-devices hang is the same single-device GBTLA recipe adopted in the Foundry I reproduction (Huang et al., 2025).

3 Results

3.1 MAP recovers the truth on all 12 systems

The MAP stage reaches the noise floor on every system: reduced χ_ν^2 ranges 1.013–1.110 (Table 1, “MAP χ_ν^2 ” column), i.e. the data are fit to within their per-pixel noise and the optimizer is not stuck on a simulator-mismatch floor. This confirms the multi-start MAP finds the global optimum from broad-prior initialization, as the paper reports for its 100 systems.

3.2 Per-system convergence: meet-target vs. shortfall

Table 1 gives, per system, the MAP χ_ν^2 and the all-22 parameter min-ESS and max- $\hat{R}$ from the 50-chain $\times$ 750-draw HMC. The systems split cleanly:

- **Meet-target (systems 4, 8, 9, 10).** Min-ESS 7,572–18,289 and max- $\hat{R}$ 1.005–1.015, both within the paper’s uniform $\hat{R} \leq 1.017$ target; system 9 even reaches a per-parameter ESS of 30,494 and $\hat{R} = 1.0005$, matching the paper’s mean-ESS regime. On these systems well-mixed parameters have $\mu_z \approx 0$ (e.g. `l1_center`, `l1_R_sersic`, pooled $\hat{R} \approx 1.01$), confirming unbiased recovery.
- **Shortfall (systems 3, 6, 7, 11).** A handful of strongly correlated mass directions (typically `center_y`, shear, source ellipticity) stall at min-ESS 71–145 and max- $\hat{R}$ up to 5.97 (system 11). These are the pooled worst case.

Table 1: Per-system results for the 12-system validation subset. MAP χ_ν^2 is the reduced chi-square at the multi-start optimum; min-ESS and max- $\hat{R}$ are over all 22 parameters from the 50-chain $\times$ 750-draw HMC; t_{HMC} is single-A16 HMC wall-clock. “Meet” = both within the paper’s uniform $\hat{R} \leq 1.017$ target.

sys	θ_E	arc SNR	MAP χ_ν^2	min-ESS	max- $\hat{R}$	t_{HMC}
0	1.290	210	1.019	419	1.305	23 min
1	1.363	116	1.026	3,891	1.028	12 min
2	1.310	155	1.013	3,784	1.028	23 min
3	2.082	453	1.110	117	2.066	21 min
4	1.062	164	1.023	7,572	1.011	meet
5	1.023	116	1.064	449	1.249	21 min
6	1.626	86	1.055	107	2.866	24 min
7	1.250	94	1.060	145	2.345	21 min
8	0.809	156	1.018	11,203	1.007	meet
9	1.023	95	1.041	10,527	1.007	meet
10	0.949	45	1.032	6,330	1.015	meet
11	1.261	63	1.104	71	5.966	25 min

3.3 Pooled statistics and the unbiasedness check

Pooling all 12 systems with `04_pool_results.py` (Table 2) reproduces the paper’s Table-2 layout. The pooled mean ESS is $\sim 11,000$. Well-mixed parameters — the lens-light position and size — have pooled $\hat{R} \approx 1.01$ –1.02 and small μ_z , exactly the unbiased behavior the paper reports. The

pooled *worst case* (min-ESS 71, max- $\hat{R}$ 5.97) falls short of the paper’s uniform min-ESS = 26,822 / max- $\hat{R} \leq 1.017$.

Crucially, the large $|\mu_z|$ values track the low-ESS/high- $\hat{R}$ parameters *exactly*: the mass ellipticity components e_1 , e_2 have the largest $|\mu_z|$ (−15.4, +16.6) *and* the highest max- $\hat{R}$ (2.21, 4.69), all driven by the single worst system (#11), where 50 chains $\times$ 750 draws have not mixed. This is the signature of a non-convergence, variance-under-estimation artifact (a too-narrow, mis-centered posterior on an unconverged chain), *not* a recovery bias: on the meet-target systems the same parameters have μ_z within $\pm 2\sigma$ of zero. The 68% / 95% truth-coverage fractions (0.63 / 0.88 pooled) are depressed by the same unconverged systems.

Table 2: Pooled statistics over the 12 systems (paper Table 2 layout), for the 8 lensing parameters. μ_z is the mean scaled error $\langle(\text{mean} - \text{truth})/\sigma\rangle$ (paper target ≈ 0); max- $\hat{R}$ / min-ESS are over all 12 systems. The paper’s uniform targets are max- $\hat{R} \leq 1.017$, min-ESS $\geq 26,822$. The large $|\mu_z|$ entries coincide with the highest $\hat{R}$ (unconverged system 11), not with a bias.

param	μ_z	$\langle\hat{R}\rangle$	max- $\hat{R}$	min-ESS
θ_E	1.97	1.176	2.375	143
γ	1.84	1.128	2.091	134
e_1	−15.35	1.236	2.210	110
e_2	16.59	1.409	4.694	75
center_x	6.30	1.324	4.058	83
center_y	−1.08	1.804	5.966	71
γ_1	−5.10	1.438	4.930	76
γ_2	0.46	1.277	2.579	96
<i>Overall (all 22 params, all 12 systems):</i> max- $\hat{R} = 5.97$, min-ESS = 71, mean-ESS = 11,082.				
<i>Truth coverage:</i> 68% = 0.63, 95% = 0.88.				

3.4 Depth test: monotone, bias-free improvement

To confirm the shortfall is a sampling-*depth* limitation and not a method or modeling defect, we re-ran the three worst systems (0, 3, 5) at $2.5\times$ depth: burn-in 500 and 1500 retained draws (vs. baseline 250 / 750), same kernel, same $\tilde{M}$ (`data/deep_run.log`). Table 3 shows the result: min-ESS roughly *doubles* and excess $\hat{R}$ roughly *halves* on every system, monotonically. System 5’s mass-only block reaches $\hat{R} = 1.007$ / min-ESS 11,821 — comfortably inside the paper’s target — at $2.5\times$ depth. The posterior means are unchanged to three decimals (e.g. system 0 θ_E mean 1.2879 $\rightarrow$ 1.2880), confirming the extra draws sharpen the variance estimate without moving the center: a bias-free convergence improvement, exactly as expected if the only deficit is chain length.

Wall-clock. On a single 15 GB A16, MAP is ≈ 75 s and SVI ≈ 217 s per system (shared compilation cache; identical kernel shapes across systems). HMC is ≈ 11 –23 min/system at baseline and ≈ 43 –49 min/system at $2.5\times$ depth. The dominant cost is the GBTLA trajectory adaptation, which at 80×80 , $k_{\text{super}} = 2$ issues up to 30 gradient evaluations per step. The paper’s 105 s/system is on a $4\times$ -A100 Perlmutter node; our per-system wall-clock is the expected ~ 10 – $20\times$ slower on a single, much weaker GPU.

Table 3: Depth test on the three worst systems. Baseline = 50 chains $\times$ 750 draws; deep = 50 chains $\times$ 1500 draws ($2.5\times$ total samples incl. longer burn-in). min-ESS / max- $\hat{R}$ over all 22 params. System 5 mass-only deep: $\hat{R} = 1.007$, min-ESS = 11,821.

sys	baseline		deep		$t_{\text{HMC}}(\text{deep})$
	min-ESS	max- $\hat{R}$	min-ESS	max- $\hat{R}$	
0	419	1.305	622	1.166	49 min
3	117	2.066	240	1.485	47 min
5	449	1.249	777	1.147	43 min

4 Caveats and scope

12 validation systems, not the full 100. We reproduce the benchmark on a 12-system subset. The split we observe (8 of 12 well-mixed, 4 with one or more stalled mass directions) is informative about the method’s behavior but does not reproduce the paper’s full Table 2, whose strength is precisely the *uniformity* of convergence across 100 systems.

Single A16 vs. $4\times$ A100. The paper ran on 40–80 GB A100s (4 per Perlmutter node); each mock here runs on one 15 GB A16. To fit the 80×80 supersampled model in memory and bound wall-time, the MAP multistart is reduced $K_{\text{MAP}}: 300\rightarrow 128$ and the SVI MC batch $K_{\text{VI}}: 1000\rightarrow 200$, and float32 is used (the paper and the canonical `jax-demo` are also float32). The model, grid, noise, priors, and the full HMC kernel stack ($\tilde{M} = \Sigma_{\text{VI}}^{-1}$, GBTLA, dual-averaging) are *unchanged*.

Uniform-ESS target not reached on the hardest systems. The paper’s min-ESS = 26,822 over all parameters and all 100 systems is a high bar that, as the depth test shows, requires more retained draws than our single-A16 wall-time budget allowed on the most degenerate posteriors. This is consistent with the Foundry I (Huang et al., 2025) finding that HMC mixing on degenerate strong-lens posteriors is the binding constraint on single-device hardware: the deficit is sampling efficiency / depth, not a defect in the GIGA-Lens model or its inference algorithm. The bias-free, monotone improvement under deeper sampling (§3.4) is the direct evidence for this attribution.

5 Conclusions

The GIGA-Lens (Gu et al., 2022) method — self-consistent EPL+shear+Sérsic mock generation followed by the MAP $\rightarrow$ SVI $\rightarrow$ preconditioned-GBTLA-HMC pipeline — is faithfully reproduced with the upstream `gigalens` JAX code. MAP recovers the injected truth at the noise floor ($\chi^2_\nu \approx 1.0$) on all 12 validation systems; on well-conditioned systems the HMC *meets* the paper’s uniform convergence target ($\hat{R} \leq 1.017$, ESS in the 10^4 range) with $\mu_z \approx 0$. The pooled shortfall (worst-case min-ESS 71, max- $\hat{R}$ 5.97) is a sampling-depth/efficiency limitation of running degenerate strong-lens posteriors on a single 15 GB A16 rather than the paper’s $4\times$ -A100 node: the large μ_z values track the unconverged parameters exactly, and a $2.5\times$ -depth re-run of the worst systems roughly doubles min-ESS and halves excess $\hat{R}$ monotonically and without moving the posterior means. This matches the Foundry I HMC-degeneracy finding and confirms the reproduction isolates a hardware/depth limit, not a method defect.

Software and data. Scripts: reproductions/gu-2022/01--05_*; gigalens source at /raid/benson/lensing-r environment /raid/benson/.venvs/gigalens (JAX, TensorFlow Probability, Optax, lenstronomy, astropy (Astropy Collaboration, 2022)). Per-system fits, logs, and the pooled / depth-test outputs are under reproductions/gu-2022/data/. Code repository: <https://github.com/usfcs-edu/agentik-lensing>.

A Side-by-side comparison with the published figures and tables

For visual verification, the panel below pairs the summary-statistics table reproduced in this report **(a, top)** with the corresponding table from the published paper **(b, bottom)**. The published table is reproduced from [Gu et al. \(2022\)](#) for comparison; the full discussion is in §3.3.

(a) This work (Table 2)

param	μ_z	$\langle \hat{R} \rangle$	max- $\hat{R}$	min-ESS
θ_E	1.97	1.176	2.375	143
γ	1.84	1.128	2.091	134
e_1	-15.35	1.236	2.210	110
e_2	16.59	1.409	4.694	75
center_x	6.30	1.324	4.058	83
center_y	-1.08	1.804	5.966	71
γ_1	-5.10	1.438	4.930	76
γ_2	0.46	1.277	2.579	96
<i>Overall (all 22 params, all 12 systems): max-$\hat{R}$ = 5.97, min-ESS = 71, mean-ESS = 11,082.</i>				
<i>Truth coverage: 68% = 0.63, 95% = 0.88.</i>				

(b) [Gu et al. \(2022\)](#), Table 2

Parameter	Mean error	μ_z	$\langle \hat{R} \rangle$	max $\hat{R}$	$\langle ESS \rangle$	min ESS
θ_E	-0.00026	-0.04 ± 0.09	1.001	1.013	35465	30846
γ_{epl}	0.01608	0.12 ± 0.08	1.001	1.017	35407	28045
$\epsilon_{epl,1}$	0.00235	0.08 ± 0.09	1.001	1.011	35590	29438
$\epsilon_{epl,2}$	-0.00159	0.01 ± 0.10	1.001	1.006	35505	31213
x_{epl}	0.00031	-0.10 ± 0.10	1.001	1.004	35617	33434
y_{epl}	0.00082	0.06 ± 0.09	1.001	1.008	35569	32745
$\gamma_{ext,1}$	0.00088	0.07 ± 0.09	1.001	1.012	35542	27926
$\gamma_{ext,2}$	-0.00060	-0.06 ± 0.09	1.001	1.010	35382	30456
R_l	-0.00203	-0.09 ± 0.09	1.000	1.007	35768	33695
n_l	-0.00321	-0.05 ± 0.08	1.000	1.008	35777	33752
$\epsilon_{l,1}$	-0.00038	-0.21 ± 0.10	1.000	1.003	35444	33593
$\epsilon_{l,2}$	-0.00006	-0.01 ± 0.10	1.001	1.003	35495	33247
x_l	0.00003	0.03 ± 0.09	1.001	1.003	35687	33086
y_l	0.00006	0.13 ± 0.08	1.000	1.003	35641	32861
I_l	1.20330	0.07 ± 0.08	1.000	1.007	35758	33603
R_s	0.00599	0.13 ± 0.09	1.000	1.005	35497	32109
n_s	0.01192	0.04 ± 0.09	1.000	1.005	35613	32820
$\epsilon_{s,1}$	-0.00255	-0.09 ± 0.10	1.000	1.004	35556	31342
$\epsilon_{s,2}$	0.00282	0.05 ± 0.10	1.001	1.005	35711	31810
x_s	-0.00087	-0.06 ± 0.09	1.001	1.017	35443	26822
y_s	-0.00092	0.03 ± 0.08	1.001	1.017	35470	29876
I_s	-0.09109	-0.05 ± 0.09	1.000	1.003	35580	33416

Figure 1: Summary statistics of the recovered lensing parameters. (a) This work: the pooled mean scaled error μ_z , $\langle \hat{R} \rangle$, max- $\hat{R}$, and min-ESS over the 12-system validation subset (report Table 2), laid out in the paper’s Table-2 format. (b) The published Table 2, the same diagnostics over all 100 simulated systems and all 22 model parameters. On the well-mixed parameters our μ_z sits near zero, matching the paper’s unbiased recovery ($\mu_z \approx 0$ throughout their Table 2); the large $|\mu_z|$ / low-ESS entries (e.g. e_1, e_2) coincide with the single unconverged system 11 — a sampling-depth artifact of the single 15 GB A16 run rather than a recovery bias (§3.3). The paper’s uniform target of min-ESS = 26,822 over all 100 systems is the high bar not reached on our most degenerate systems; note that the published table tabulates all 22 parameters at near-uniform $\langle \hat{R} \rangle = 1.000$ –1.001, whereas we pool the 8 lensing parameters. Panel (b) reproduced from [Gu et al. \(2022\)](#), their Table 2, for comparison.

References

- Astropy Collaboration. The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935:167, 2022. doi: 10.3847/1538-4357/ac7c74.
- S. Birrer, A. Shajib, D. Gilman, A. Galan, J. Aalbers, M. Millon, R. Morgan, G. Pagano, J. W. Park, L. Teodori, et al. lenstronomy II: A gravitational lensing software ecosystem. *Journal of Open Source Software*, 6(62):3283, 2021. doi: 10.21105/joss.03283.
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Neca, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs. <http://github.com/google/jax>, 2018.
- J. V. Dillon, I. Langmore, D. Tran, E. Brevdo, S. Vasudevan, D. Moore, B. Patton, A. Alemi, M. Hoffman, and R. A. Saurous. TensorFlow Distributions. *arXiv preprint*, 2017.
- A. Gu, X. Huang, W. Sheu, G. Aldering, A. S. Bolton, K. Boone, A. Dey, A. Filipp, E. Jullo, S. Perlmutter, D. Rubin, E. F. Schlafly, D. J. Schlegel, Y. Shu, and S. H. Suyu. GIGA-Lens: Fast Bayesian Inference for Strong Gravitational Lens Modeling. *The Astrophysical Journal*, 935(1):49, 2022. doi: 10.3847/1538-4357/ac6de4.
- M. D. Hoffman and A. Gelman. The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15:1593, 2014.
- M. D. Hoffman, A. Radul, and P. Sountsov. An Adaptive-MCMC Scheme for Setting Trajectory Lengths in Hamiltonian Monte Carlo. *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 130:3907, 2021.
- X. Huang, C. Storfer, A. Gu, V. Ravi, A. Pilon, W. Sheu, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.
- X. Huang, S. Baltasar, N. Ratier-Werbin, et al. DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens. *arXiv preprint*, 2025.
- N. Tessore and R. B. Metcalf. The elliptical power law profile lens. *Astronomy & Astrophysics*, 580: A79, 2015. doi: 10.1051/0004-6361/201526773.